



Exam for 2nd year Biochemistry students

Answer the following questions:

Section (1): Carbohydrates

A- Choose the correct answer:

(16 Marks)

1- Glucose and fructose have the same configuration in C atoms number:

- A) 1, 2 B) 2,3,4,5 C) 3,4,5,6 D) 4,5,6

2- The following examples are important heteropolysaccharides except

- A) Amylopectin B) Heparin C) Hyaluronic acid D) Peptidoglycan

3- The glycosaminoglycan which does not contain uronic acid is

- A) Dermatan sulphate B) Keratan sulphate C) Heparan sulphate D) Chondroitin sulphate

4- α -D-Glucuronic acid is present in

- A) Chondroitin sulphate B) Heparin C) Hyaluronic acid D) All of these

5- A carbohydrate, known commonly as invert sugar, is

- A) Fructose B) Sucrose C) Glucose D) Lactose

6- Glucopyranose has free - OH groups number:

- A) 1,2,3,4,6 B) 1,2,5,6 C) 2,3,4,5,6 D) 1,2,3,5,6

7- A polysaccharide which is often called animal starch is:

- A) Glycogen B) Inulin C) Starch D) Dextrin

8- A disaccharide formed by 1,1-glycosidic linkage between their monosaccharide units is:

- A) Maltose B) Cellobiose C) Lactose D) Trehalose

9- Hyaluronic Acid is unique among GAGs because it:

- A) Does not contain any sulfate groups B) Lacks a uronic acid sugar C) Is highly sulfated D) Is covalently linked to a core protein

10- Heparin exerts its anticoagulant effect primarily by binding to and activating:

- A) Antithrombin III B) Fibrinogen C) Vitamin K D) Calcium ions

11- The structural polysaccharide found in the exoskeleton of insects and the cell walls of fungi is:

- A) Chitin B) Pectin C) Amylose D) Inulin

12- Which glycosidic linkage connects the two galactose units in Stachyose?

- A) α (1-2) B) α (1-6) C) α (1-4) D) β (1-4)

13- In the structure of Lactose, which sugar provides the anomeric carbon for the glycosidic bond?

- A) Glucose B) Galactose C) Fructose D) Mannose

14- The heteropolysaccharide 'Agar' is widely used in microbiology. It is primarily obtained from:

- A) Arthropod exoskeletons B) Seaweed C) Bacterial fermentation D) Connective tissue

15- Cellobiose is obtained as a partial hydrolysis product of which polysaccharide?

- A) Glycogen B) Cellulose C) Starch D) Chitin

16- Inulin is a homopolysaccharide composed of repeating units of:

- A) Glucose B) Galactose C) Fructose D) Mannose

B- Please, put T for true and F for False for each of the following: (10 marks)

- | | | |
|----|--|---------|
| 1 | An acetal of the pyranose or furanose form of a sugar is referred to as a glycoside. | () |
| 2 | A monosaccharide can contain the carbonyl of a ketone or the carbonyl of an aldehyde | () |
| 3 | Starch, glycogen, and cellulose are all examples of oligosaccharides. | () |
| 4 | An L-sugar and a D-sugar of the same name are enantiomers. | () |
| 5 | Aldonic are oxidized carbohydrates. | () |
| 6 | D-Glucose and D-ribose are diastereomers. | () |
| 7 | A pyranoside contains a five-membered ring. | () |
| 8 | Polysaccharide is a glycoside of two monosaccharides. | () |
| 9 | The penultimate carbon of an acyclic monosaccharide becomes the anomeric carbon in the cyclic hemiacetal form of the molecule. | () |
| 10 | Mutarotation is the establishment of an equilibrium concentration of α and β anomers of a carbohydrate. | () |

C- What is the main structural and importance difference between? (15 marks)

- a- Lactose and Maltose.
b- Agar, agarose and pectins.
c- glucose and galactose.

D- The anticlotting property of heparin is partly the result of the negative charges it carries.(9 marks)

- (a) draw the structure of heparin.
(b) Identify the functional groups that provide the negative charges.
(c) Describe the glycosidic bond in heparin.

E- Explain how the carbohydrate content plays a determinant role in blood grouping (5 marks)

F- Define the following with examples: (5 marks)

Anomers - Epimers

Section (2): Lipids

A- Give the chemical structures of the following fatty acid: (12 Marks)

- 1.) 16:0 2.) 18:2 $\Delta^{3,6}$ 3.) 20:4 $\Delta^{5,9,12,15}$ 4.) 20:1(ω -6)

B- Classify lipids according to simple and complex. (8 Marks)

C- Illustrate with examples the classification of fatty acids. (8 Marks)

D- Summarize the properties of fats and oils. (8 Marks)

E- Point out the properties of cell membranes. (8 Marks)

F- Define terpenes and memorize their classification and function. (8 Marks)

G- Compare between soaps and detergents. (8 Marks)

All the best.....

Dr. Medhat Owda

Dr. Abdelaaty Hamed



كيمياء حيوية عام (١) - ك ج ٢٦١
(ثانية كيمياء حيوية)

Al-Azhar University
Faculty of Science "Boys"
Chemistry Department

Academic Year 2024/2025 (First Term)
Final Examination For Second Year, Biochemistry Students.
Course Title: "General Biochemistry (1)" Course Number : Bio. Chem. 261

January 2025

Time allowed : 3 hours

Section A: Lipids (60 Marks)

1. A) Give names and chemical structures of **hydrolysis products** of the following: (10 marks)
Lecithins , Sphingomyelins and Galactolipids .
- B) Explain briefly the following methods of analysis of Fats and Oils : (10 marks)
Acid Value , Saponification Value and Iodine Value.
- C) Draw the chemical structure of the following: (10 marks)
> Testosterone > Estradiol > Cortisone > Low Density Cholesterol (LDL)
> Androsterone > Progesterone > Vitamin D₂ > High Density Cholesterol (HDL)
2. A) Explain by chemical equations , How can you prepare the following **Surfactants**: (12 marks)
i. Sodium Lauryl ether sulphate . From Lauric acid
ii. Sodium palmityl glycerol ether sulphonate . From Palmitic acid .
iii. Diethanol oleyl amine hydrochloride . From Oleic acid.
iv. Betaines derived from Stearyl imidazoline derivatives. From Stearic acid.
- B) Discuss the **addition reactions** of the hydrocarbon chain of fatty acids. (10 marks)
- C) Discuss what do you know about **TWO ONLY** of the following? (8 marks)
i. Biological functions of fats and oils .
ii. Properties and function of phospholipids
iii. Cell Membrane properties.
iv. Physiological importance of waxes.

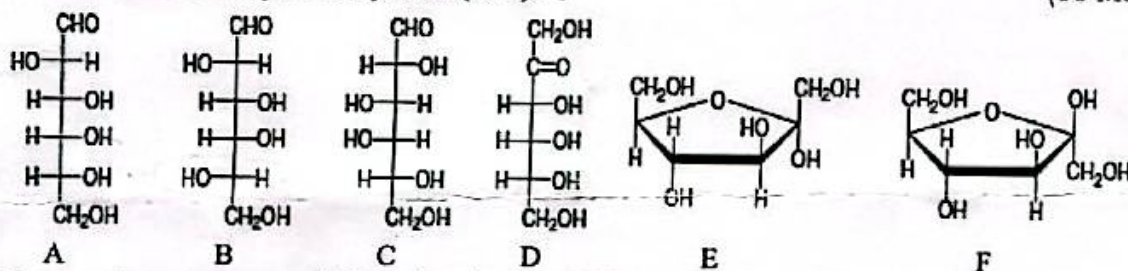
Section B: Carbohydrates (60 Marks)

- 3.A) Choose the correct answer: - (20 Marks)
- A tetrose sugar is
(a) Dihydroxyacetone (b) Ribulose (c) Erythrose (d) Glucose
 - A carbohydrate found only in milk is
(a) Maltose (b) Sucrose (c) Lactose (d) Galactose
 - Cellulose is made up of the molecules of
(a) α -glucose (b) β -glucose (c) α -galactose (d) None of these
 - Mutarotation refers to change in
(a) Chemical properties (b) pH (c) Conductance (d) Optical rotation
 - Fructose is present in hydrolysate of
(a) Starch (b) Inulin (c) Lactose (d) None of these
 - A disaccharide linked by α -1-4 Glycosidic linkages is
(a) Lactose (b) Sucrose (c) Cellulose (d) Maltose
 - Which of the following is a heteroglycan?
(a) Inulin (b) Dextrin (c) Chitin (d) Agar
 - Maltose can be formed by hydrolysis of
(a) Starch (b) Dextrin (c) Glycogen (d) All of these
 - A carbohydrate, known commonly as invert sugar, is
(a) Fructose (b) Sucrose (c) Glucose (d) Lactose
 - The carbohydrate of the blood group substances is
(a) Sucrose (b) Fucose (c) Arabinose (d) Maltose

11. The polysaccharide found in the exoskeleton of invertebrates is
(a) Pectin (b) Cellulose (c) Chitin (d) None of these
12. A polysaccharide which is often called animal starch is
(a) Glycogen (b) Inulin (c) Starch (d) Dextrin
13. In place of glucuronic acid chondroitin sulphate contains
(a) Gluconic acid (b) Galactose (c) Iduronic acid (d) Sulphonic acid
14. A disaccharide formed by 1,1-glycosidic linkage between their monosaccharide units is
(a) Maltose (b) Cellobiose (c) Lactose (d) Trehalose
15. Hyaluronic acid is found in
(a) Joints (b) Brain (c) Abdomen (d) Mouth
16. The glycosaminoglycan which does not contain uronic acid is
(a) Dermatan sulphate (b) Chondroitin sulphate (c) Keratan sulphate (d) Heparan sulphate
17. Keratan sulphate is found in abundance in
(a) Heart muscle (b) Liver (c) Adrenal cortex (d) Cornea
18. α -D-Glucuronic acid is present in
(a) Chondroitin sulphate (b) Heparin (c) Hyaluronic acid (d) All of these
19. The component of cartilage and cornea is
(a) Chondroitin sulphate (b) Heparin (c) Hyaluronic acid (d) Keratan sulphate
20. A branch component of starch is
(a) Amylopectin (b) Glucose (c) Amylose (d) Maltose

3.B) Consider the following six compounds (A-F).

(10 Marks)



How are the two compounds in each pair related? Choose from enantiomers, anomer, diastereomers but not epimers, Aldose-ketose isomers, and epimer compounds.

- a) A and B b) A and C c) B and C d) A and D e) E and F

4.A) a- Discuss the biological Importance's of carbohydrates

(5 Marks)

b- Write on the structure and importance of lactose.

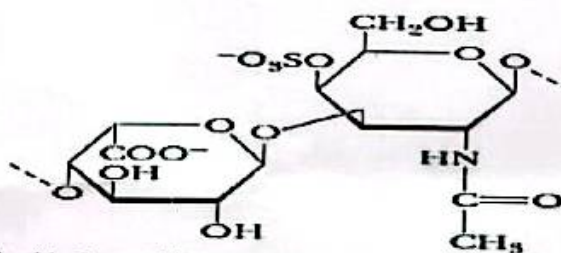
(5 Marks)

c- Write on blood group substance

(5 Marks)

d- Following is a structural formula for the repeating disaccharide unit of a heteropolysaccharide which is a component of the extracellular matrix of the skin.

(5 Marks)



(a) Name the monosaccharide from which each unit of this disaccharide is derived.

(b) Identify a heteropolysaccharide and explain its functions.

(c) Describe the glycoside bonds

4.B)

Compare between the following:

a- Cellulose and starch

(5 Marks)

b- Glucose and fructose

(5 Marks)

GOOD LUCK

أ.د. محمد مصطفى السيد مصطفى - د. مدحت السيد عيد عوده



Section A: Lipids (60 Marks)

Question1

(30 marks)

A: Compare between TWO ONLY the following

(20 Marks)

- 1) Soaps and detergents
- 2) The ABO blood group antigens
- 3) Saturated vs Unsaturated fatty acid

B: Complete the following sentences with the best answer

(10 Marks)

1- Phospholipids belong to ----- class according to functional classification of lipid

A- Storage	B -Structure	C-Function	D- Closed
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2- Accumulation of ganglioside ----- in the brain of infants leads to Tay Sachs Disease

A- GM1	B - GM2	C- GM3	D- GM4
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3- Microbial enzymes causing ----- reaction responsible for rancidity

A- Oxidation	B- Reduction	C - Hydrolysis	D- Nitration
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4- Breakdown of fat globules by ----- help in the absorption of dietary lipids

A- Cholesterol	B- Adrenal H	C - Bile salts	D- Vitamin D
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5- Eicosanoid member ----- acts as platelet aggregators

A- Prostaglandins	B - Thromboxanes	C- Leukotrienes	D- Arachidonic
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6- ----- fatty acids have been linked to increased insulin resistance and type 2 diabetes

A- Saturated	B- Omega	C -Trans	D- Cis
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7- Bee waxes composed of long-chain alcohols such as -----

A- Cetyl	B - Myricyl	C- Glycerol	D- Sphingosine
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8- Humans with B-type blood contain B enzyme adds ----- to O antigen

A- Glucose	B- Fucose	C - Galactose	D- N-Acetylgalactosamine
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9- The major apolipoproteins of HDL is ----

A- B100	B - A	C- B48	D- E
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10- ----- physiological function is a protective agent on the surfaces of animals

A- Fatty Acids	B - Waxes	C- Steroids	D- lecithin
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Question2:

(30 marks)

A: Write what you know on TWO ONLY the following

(20 Marks)

- 1) Biological functions of fats and oils
- 2) The formation and properties of cell membrane
- 3) Best known and most abundant steroid in the human body

B: Put true or false on the following:

(5 Marks)

- 1) Phosphatidylcholine is abundant in all cell extracts especially in liver and kidney (X)
- 2) Cardiolipin seems to be a potential factor in thyroid disease (✓)
- 3) Lysophosphatidic acid has several biological properties including cell division (✓)
- 4) All steroids aid in the breakdown and absorption of dietary lipids (X)
- 5) Cerebrosides glycolipid contains N-acetylneuraminic acid (NANA) (X)

C: Matches between column A with the best answer in column B

(5 Marks)

A	B
1. Sphingomyelin belong to	a) Omega 6 Fatty acid
2. $\text{CH}_3(\text{CH}_2)_4\text{C}=\text{CCH}_2\text{C}=\text{C}(\text{CH}_2)_7\text{COOH}$ is	b) Ceramide
3. Precursor of glycolipids is	c) Waxes
4. $\text{CH}_3(\text{CH}_2)_{14}\text{COOC}_3\text{H}_6\text{I}$ is	d) Steroids
5. Isoprene precursor of	e) Glycolipids
	f) Arachidonic Acid
	g) Sphingophospholipid
	h) Soap
	i) Phosphatidic acid
	j) Terpenes

Section B: Carbohydrates (60 Marks)

(20 marks)

1- Choose the correct answer: -

1. A pentulose sugar is
(a) Dihydroxyacetone (b) Ribulose (c) Erythrose (d) Glucose
2. Number of stereoisomers of glucose is ~~16~~ 16
(a) 4 (b) 8 (c) 16 (d) None of these
3. The polysaccharide found in the exoskeleton of invertebrates is
(a) Pectin (b) Chitin (c) Glycogen (d) Cellulose
4. α -D-mannose and β -D-mannose are
(a) Stereoisomers (b) Epimers (c) Keto-aldo pairs (d) Anomers
5. Fructose is present in hydrolysate of
(a) Starch (b) Inulin (c) Lactose (d) None of these
6. A disaccharide linked by α -1-4 Glycosidic linkages is
(a) Lactose (b) Sucrose (c) Cellulose (d) Maltose
7. Which of the following is a non-reducing sugar?
(a) Maltose (b) Isomaltose (c) Lactose (d) Trehalose
8. Maltose can be formed by hydrolysis of
(a) Starch (b) Dextrin (c) Glycogen (d) All of these
9. A carbohydrate, known commonly as invert sugar, is
(a) Fructose (b) Sucrose (c) Glucose (d) Lactose
10. N-acetylneuraminic acid is an example of
(a) Sialic acid (b) Mucic acid (c) Glucuronic acid (d) Hippuric acid
11. Mucopolysaccharides are
(a) Homopolysaccharides (b) Proteins (c) Heteropolysaccharides (d) Amino acids

12. The following examples are important heteropolysaccharides except
~~(a)~~ Amylopectin (b) Heparin (c) Hyaluronic acid (d) Peptidoglycan
13. In place of glucuronic acid chondroitin sulphate B contains Dermatan
 (a) Gluconic acid (b) Gulonic acid (c) ~~Iduronic acid~~ (d) Sulphonic acid
14. A heteropolysacchraide among the following is
 (a) Inulin (b) Cellulose ~~(c) Heparin~~ (d) Dextrin
15. Hyaluronic acid is found in
~~(a)~~ Joints (b) Brain (c) Abdomen (d) Mouth
16. The glycosaminoglycan which does not contain uronic acid is
 (a) Dermatan sulphate (b) Chondroitin sulphate (c) Keratan sulphate (d) Heparan sulphate
17. Keratan sulphate is found in abundance in
 (a) Heart muscle (b) Liver (c) Adrenal cortex ~~(d) Cornea~~
18. α -D-Glucuronic acid is present in
 (a) Chondroitin sulphate ~~(b) Heparin~~ (c) Hyaluronic acid (d) All of these
19. Synovial fluid contains
 (a) Chondroitin sulphate (b) Heparin ~~(c) Hyaluronic acid~~ (d) Keratin sulphate
20. The carbohydrate of the blood group substances is
 (a) Sucrose ~~(b) Fucose~~ (c) Arabinose (d) Maltose

2- Discuss the chemical structure and functions of hyaluronic acid and chondroitin Sulphate (20marks)

3- Compare between the following: (20marks)
 a- Glucose and fructose
 b- Lactose and sucrose
 c- Cellulose and starch

مع أطيب التمنيات بالنجاح



Section A: Lipids (60 Marks)

Question 1

A: Write what you know on Two only of the following

(30 marks)

(20 Marks)

- 1) Cell membrane structure and function
- 2) Structure and function of Omega Fatty acids
- 3) General structure and Biological functions of waxes

B: Complete the following sentences with the best answer (5 Marks)

1- Lipids are important dietary constituents because of their -----

A- Low energy value	B- Water soluble vitamin	C- Essential fatty acids	D- Essential amino acids
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2- Accumulation of ganglioside ----- in the brain of infants leads to Tay Sachs Disease

A- GM1	B- GM2	C- GM3	D- GM4
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3- ----- physiological function is a protective agent on the surfaces of animals

A- Fatty Acids	B- Waxes	C- Steroids	D- lecithin
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4- Breakdown of fat globules by ----- help in the absorption of dietary lipids

A- Cholesterol	B- Adrenal H	C- Bile salts	D- Vitamin D
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5- Eicosanoid member ----- that acts as platelet aggregators

A- Prostaglandins	B- Thromboxanes	C- Leukotrienes	D- Arachidonic
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C: Matches between column A with the best answer in column B (5 Marks)

A	B
1. $\text{CH}_3(\text{CH}_2)_5\text{C}=\text{C}(\text{CH}_2)_7\text{COOH}$ is	a) Shingophosholipid
2. Vitamin D is	b) Saturated fatty acid
3. Glucosylcerebroside is	c) Antioxidant
4. $\text{CH}_3(\text{CH}_2)_{14}\text{COOC}_{30}\text{H}_{61}$ is	d) Triglyceride
5. (Phosphatidylcholine) is	e) Glycerophosholipid
	f) Steroid
	g) Sphingomyelin
	h) Glycolipid
	i) Un-Saturated fatty acid
	j) Wax

(30 marks)

Question 2:

A: Compare between Two only of the following

(20Marks)

- 1) Soaps and detergents
- 2) Saturated and Unsaturated Fatty Acids
- 3) Glycerophospholipids and Glycolipids

B: Put true or false on the following:

(10 Marks)

1. Simple lipids are esters of fatty acids with glycerol alcohol ()
2. Trans fatty acid make the blood less sticky and less likely to clot ()
3. Healthy skin and hair are maintained by waxes ()
4. Glycolipids play a role in blood coagulation (X) ()
5. Patient with blood group O can only receive blood from A blood donor ()
6. Phosphatidylcholine is abundant in all cell extracts especially in liver and kidney () ✓
7. Cardiolipin seems to be a potential factor in thyroid disease ()
8. Lysophosphatidic acid has several biological effect on cell division ()
9. All steroids aid in the breakdown and absorption of dietary lipids ()
10. Cerebrosides glycolipid contains N-acetylneuraminic acid (NANA) ()

Section B: Carbohydrates (60 Marks)

(20 Marks)

1A- Choose the correct answer: -

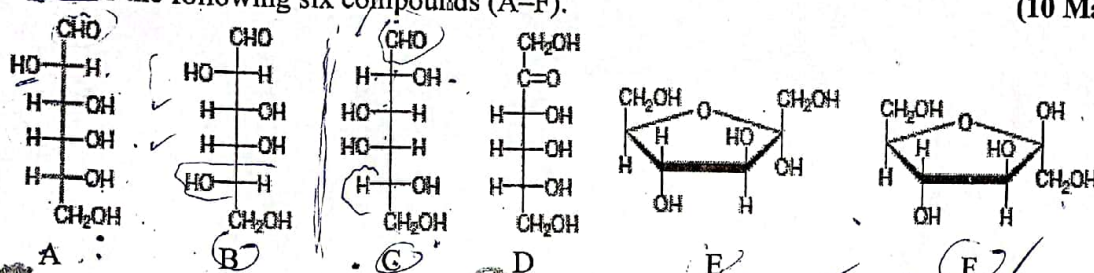
1. A pentose sugar is
(a) Dihydroxyacetone (b) Ribulose (c) Erythrose (d) Glucose
2. Number of stereoisomers of fructose is
(a) 4 (b) 8 (c) 16 (d) None of these
3. Glucose on reduction with sodium borohydride forms
(a) Mannitol and sorbitol (b) Sorbitol (c) Mannitol (d) Dulcitol
4. α -D-galactose and β -D-galactose are
(a) Stereoisomers (b) Epimers (c) Keto-aldo pairs (d) Anomers
5. Fructose is present in hydrolysate of
(a) Starch (b) Inulin (c) Lactose (d) None of these
6. A disaccharide linked by α 1-4 Glycosidic linkages is
(a) Lactose (b) Sucrose (c) Cellulose (d) Maltose
7. A carbohydrate found in DNA is
(a) Deoxyribose (b) Ribose (c) Ribulose (d) All of these
8. Maltose can be formed by hydrolysis of
(a) Starch (b) Dextrin (c) Glycogen (d) All of these
9. A carbohydrate, known commonly as invert sugar, is
(a) Fructose (b) Sucrose (c) Glucose (d) Lactose
10. N-acetylneuraminic acid is an example of
(a) Sialic acid (b) Mucic acid (c) Glucuronic acid (d) Hippuric acid
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14. A heteropolysaccharide among the following is
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9/14
- 9/14

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20. A branch component of starch is
 (a) Amylopectin (b) Glucose (c) Amylose (d) Maltose

1B- Consider the following six compounds (A-F).

(10 Marks)



How are the two compounds in each pair related? Choose from enantiomers, anomer, diastereomers but not epimers, Aldose-ketose isomers, and epimer compounds.

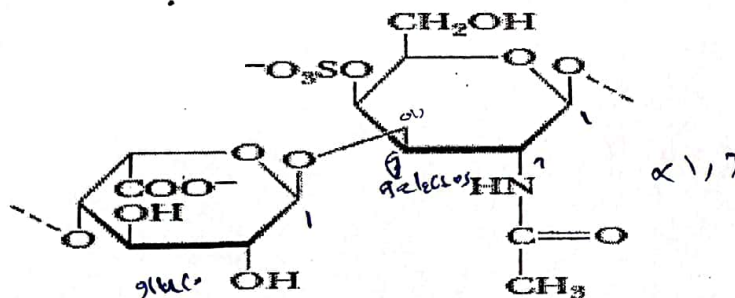
- a) A and B b) A and C c) B and C d) A and D e) E and F

2A- Starch can be separated into two principal polysaccharides: amylose and amylopectin. What is the major difference in structure between the two?

(10 Marks)

2B- Following is a structural formula for the repeating disaccharide unit of dermatan sulfate. Dermatan sulfate is a component of the extracellular matrix of the skin.

(10 Marks)



- (a) Name the monosaccharide from which each unit of this disaccharide is derived.
 (b) Describe the glycoside bonds in dermatan sulfate.

2C-

Compare between the following:

(10 Marks)

- a- Hyaluronic acid and heparin
 b- Lactose and maltose
 c- Glucose and fructose

مع أطيب التمنيات بالنجاح